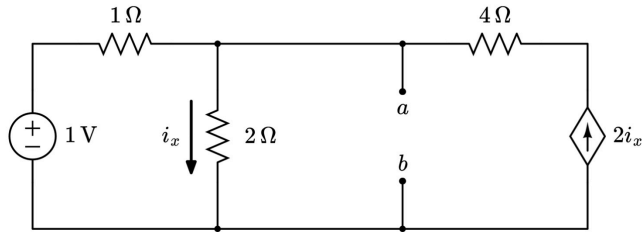


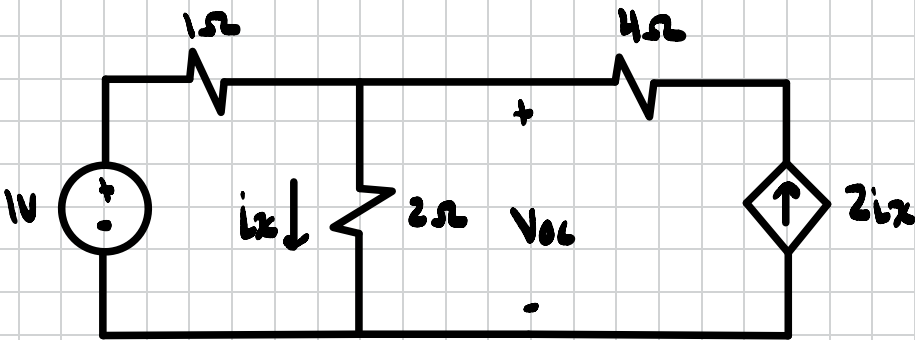
For the circuit shown below:

- Find the Thévenin equivalent circuit with respect to the open terminal pair ab .
- Find the Norton equivalent circuit with respect to the open terminal pair ab .
- Clearly illustrate both equivalent circuits with every value labeled.
- If a load is connected to points a and b , what is the maximum power that can be delivered to a load resistor R_L ?
- Determine the value of the load resistor R_L necessary to achieve maximum power transfer.

Suggested time limit: 20 minutes



Voc Method



Step #1 Constraint

$$i_x = \frac{V_{oc}}{2}$$

Step #2 Nodal @ A

$$\frac{V_{oc} - 0}{2} + \frac{V_{oc} - 1}{1} - 2i_x = 0$$

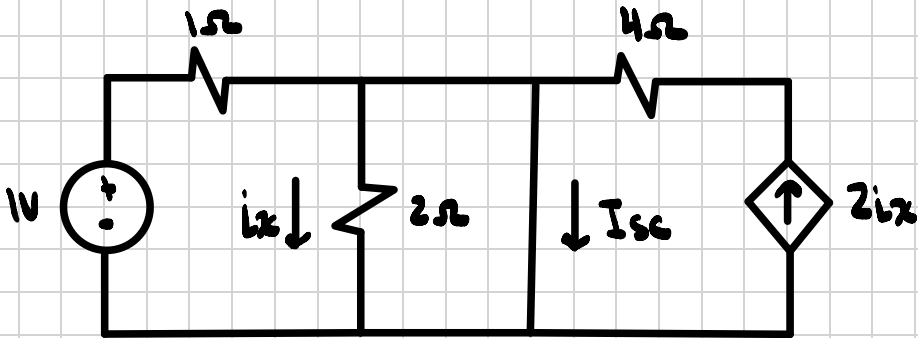
$$\frac{V_{oc}}{2} + V_{oc} - 1 - 2\left(\frac{V_{oc}}{2}\right) = 0$$

$$\left(\frac{V_{oc}}{2} + V_{oc} - 1 - V_{oc} = 0\right) 2$$

$$V_{oc} + 2V_{oc} - 2 - 2V_{oc} = 0$$

$$V_{oc} = 2V = V_{TH}$$

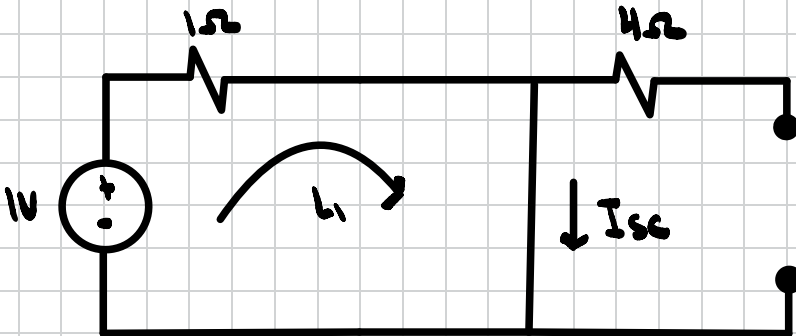
Isc Method



Step #1 Constraint

2Ω shorts $i_x = 0A$ so dependent source is $0A$ so source can be replaced with an open circuit

Step #2 Redraw Circuit

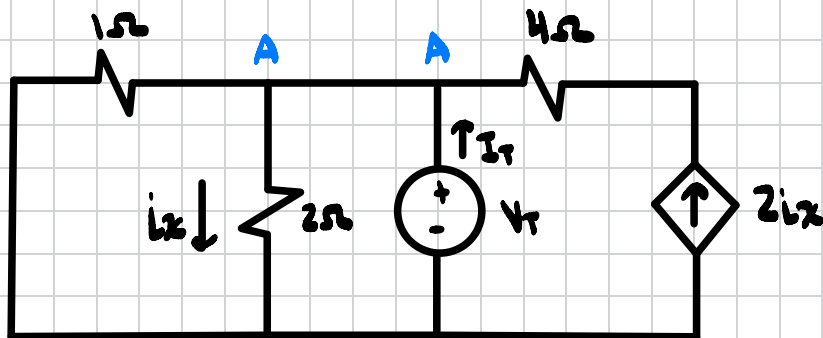


Step #3 KVL @ L_1

$$-1 + I_{sc} = 0$$

$$I_{sc} = 1A = I_N$$

Test Voltage Method



Step #1 Constraint

$$i_x = \frac{V_T}{2}$$

Turn off all
independent source
Voltage Source Short
Current Source Open

Step #2 Nodal A

$$\frac{V_T - 0}{1} + \frac{V_T - 0}{2} - I_T - 2i_x = 0$$

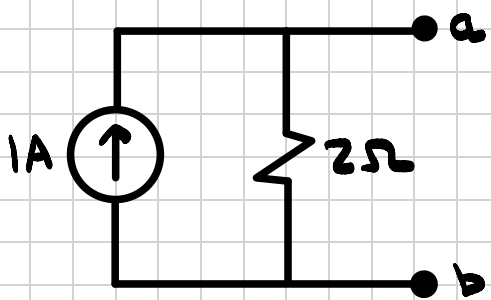
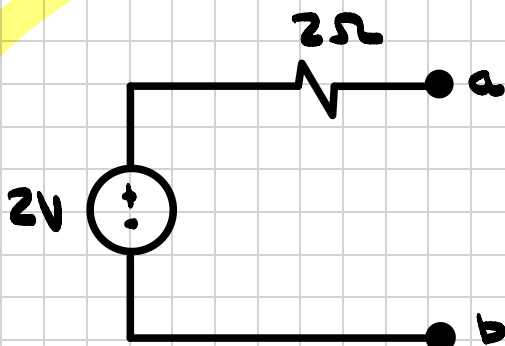
$$V_T + \frac{V_T}{2} - I_T - 2\left(\frac{V_T}{2}\right) = 0$$

$$\left(V_T + \frac{V_T}{2} - I_T - V_T = 0\right) 2$$

$$2V_T + V_T - 2I_T - 2V_T = 0$$

$$V_T = 2I_T$$

$$\frac{V_T}{I_T} = 2\Omega = R_{TH}$$



Max Power Transfer

Step #1 Formula

$$P_{MAX} = \frac{(V_{TH})^2}{4R_{TH}}$$

$$R_L = R_{TH}$$

Step #2 Solve

$$P_{MAX} = \frac{(2)^2}{4(2)} = 0.5W$$